



Applied AI & Business Capstone Design

Fall 2026 Project Pool

Inha University in partnership with
HumbleBeeAI

Course Rules & Goal

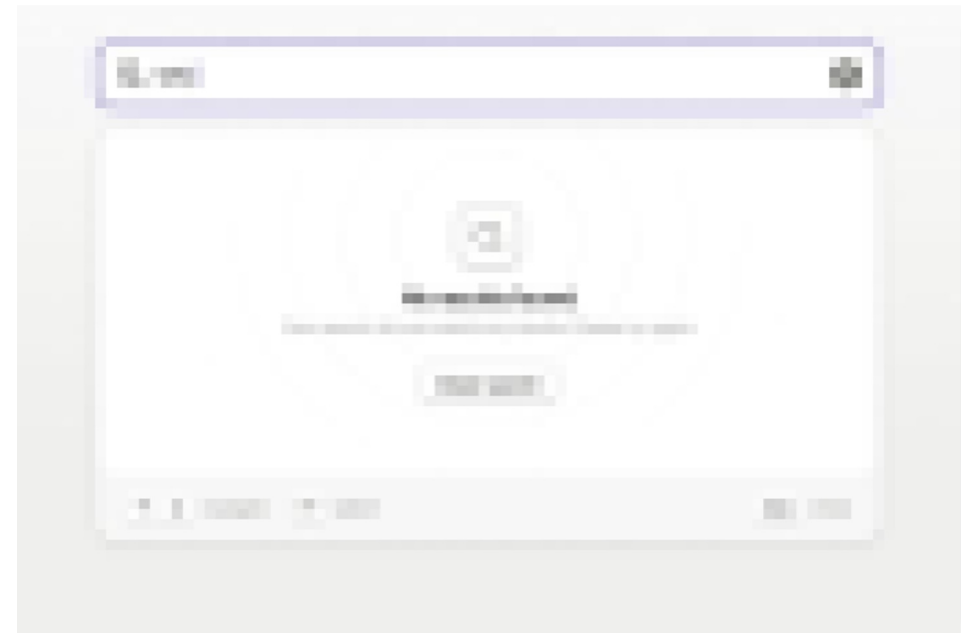
- **The Goal:** Deliver a working AI prototype or verifiable workflow demo. Slides alone do not replace code.
- **Prerequisites:** Basic Python is enough to reach the baseline for every project.
- **Data Rule:** Use public educational or safe sample data. No personal, real customer, or internal company data.



N1: Multi-Source Knowledge Recovery

The Problem

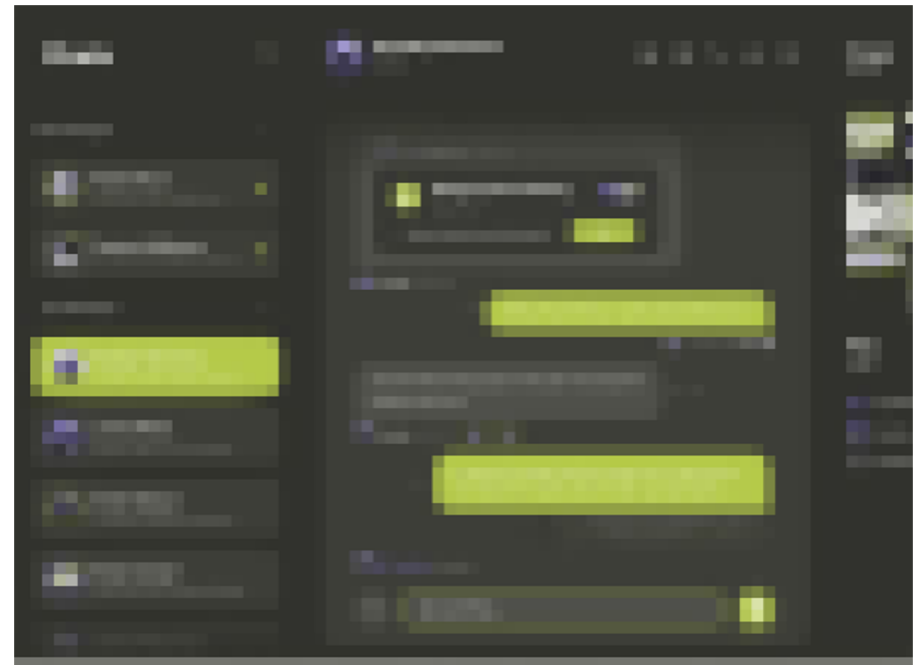
- **The Core Issue:** Organizations pay staff to answer questions that were already answered months ago.
- **Real Example:** A student asks an error question. The answer exists in a 4-month-old Discord thread, but standard search misses it due to varied phrasing.
- **Target User:** Community or Support Managers trying to deflect repeated tickets.



N1: The Solution & Hard Parts

What to Build

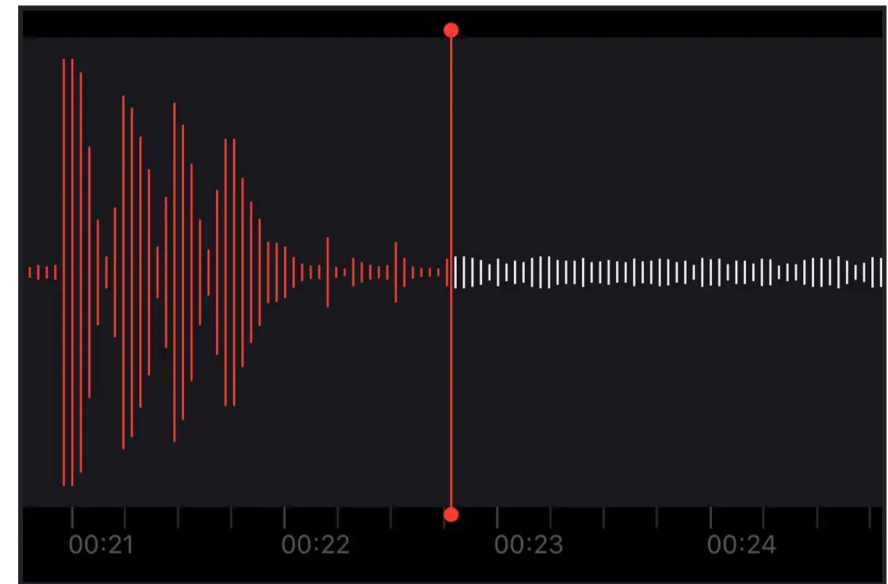
- **The Solution:** Ingest message history into a searchable store and answer questions with links to the source messages.
- **Why AI?:** Simple search fails on varied phrasing. AI handles retrieval and summarization together.
- **Known Hard Part:** Thread structures are messy, and old answers go stale quickly.



N2: Code-Switched Voice Agent

The Problem

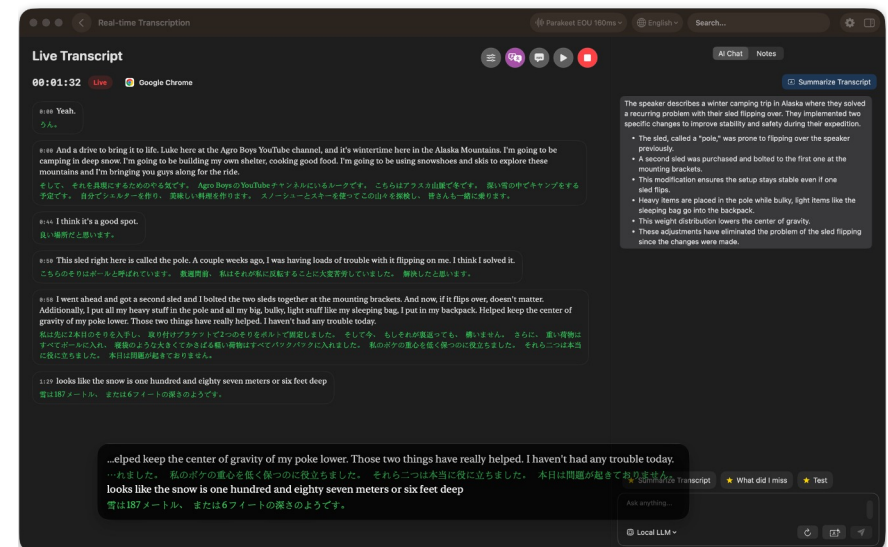
- **The Core Issue:** Voice agents work in English but fail badly in markets where people mix languages.
- **Real Example:** A user says, "*Book a table for two at five pm, please — actually, wait, make it six.*" Standard voice agents get confused by mid-sentence corrections and conversational shifts.
- **Target User:** Clinics or service businesses in non-English markets with high call volumes.



N2: The Solution & Hard Parts

What to Build

- **The Solution:** A voice pipeline (speech in → AI → speech out) tested on a custom code-switched dataset you will record.
- **Your Deliverable:** Measure the "Word Error Rate" gap between monolingual and mixed speech.
- **Known Hard Part:** System latency (delay) between modules is usually the bottleneck, not the model itself.



C1: Visual Document Understanding

The Problem

- **The Core Issue:** Standard OCR extracts text but completely destroys the document's layout.
- **Real Example:** An invoice has a neat table. Traditional OCR turns the table into a long, useless string of numbers where columns are lost.
- **Target User:** Finance or logistics teams doing manual data entry.

```
TERMINAL
(venv) ➔ ocr python3 index.py
ACME Bank
Credit Card
Limit Application

Limit Increase
o0 Limit Decrease

Limit/Month Requested €

Term Requested
15,000.00 18 months
Applicant Information
First Name Email Company Name
Joanna R. jo.simonitt189@gmail.com Rogers Food Stores
Last Name Website VAT number
Simonitt www.simonitt1.com 08909 9999 73
Date of Birth Social Security Number Credit Card Number IBAN
05/10/1989 449-09-0000 5500 0000 0004 0004 G882BANC20032619959333
Phone Address City ZIP Code
+44 78 7953 0927 39 Worthy Lane MAWLA TR16 2JB
Vehicle ownership VIN number Gross Annual Income € Net Worth €
No / 328,000.00 3,020,020.00

As reported on last Tax Return

Excluding busine:

Signature of Applicant

Application Agreement

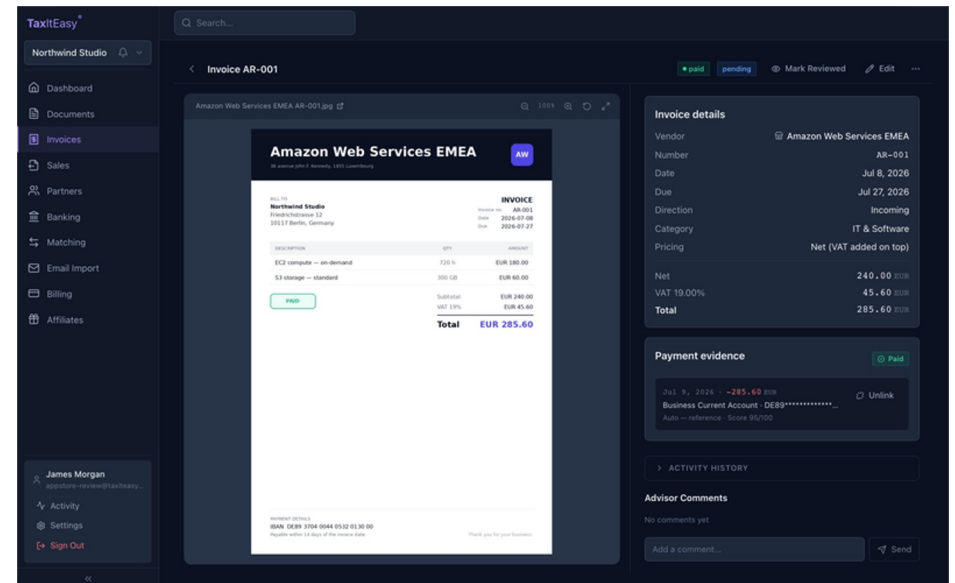
Each Signer has read and agrees to all applicable provisions on the reverse
side of this Application (whether or not that reverse side is faxed back to
ACME Bank), including the granting of a security interest in deposit accounts,
and understands that this Application may be approved or denied. ACME
Bank is authorized to obtain credit records and other credit and employment
information about the Signers personally (now and in the future), including
from state and federal tax authorities, for deciding whether to approve the
requested limit.

Date of Signature
05/25/2022
Type of Signature
```

C1: The Solution & Hard Parts

What to Build

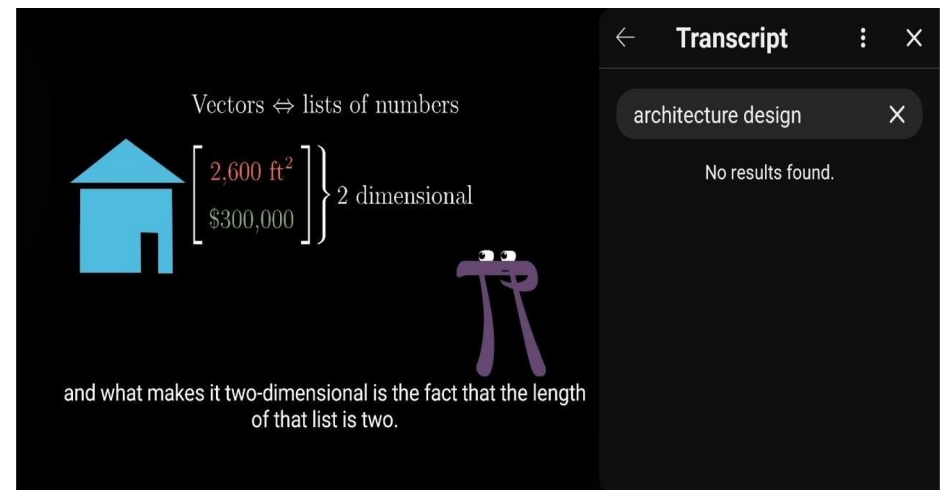
- **The Solution:** Extract fields across document types using a Vision Language Model (VLM) that "looks" at the page structure.
- **Your Deliverable:** Compare the VLM's accuracy side-by-side against a standard text-first OCR baseline.
- **Known Hard Part:** Creating manual, hand-checked ground truth data for tables is incredibly slow.



C2a: Video Search

The Problem

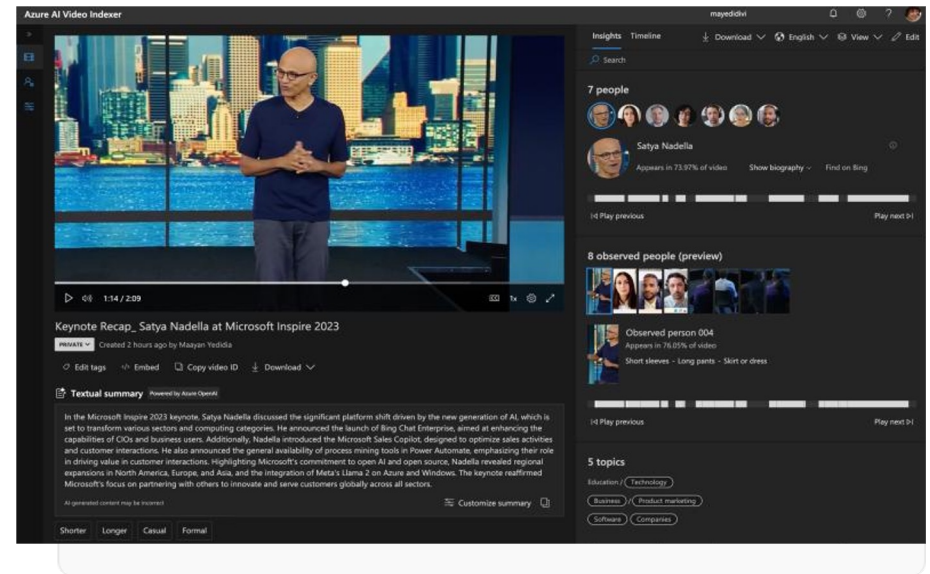
- **The Core Issue:** Video libraries are unsearchable. Transcripts only search what was *said*, ignoring what was *shown*.
- **Real Example:** A crucial architecture diagram is shown on a slide, but the speaker never says "architecture." Transcript search misses it entirely.
- **Target User:** Learning & Development teams.



C2a: The Solution & Hard Parts

What to Build

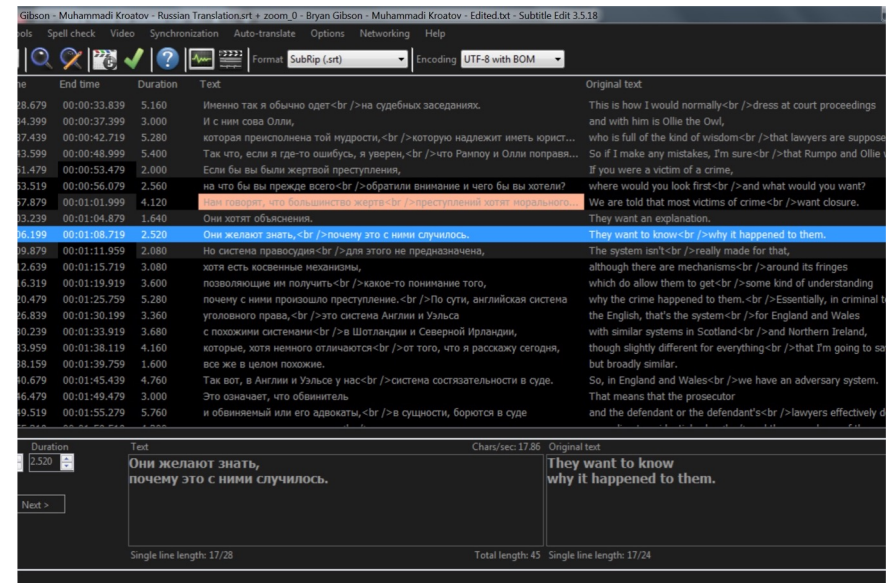
- **The Solution:** Segment videos, generate visual scene descriptions, and return timestamped frame previews for queries.
- **Why AI?:** Making video searchable requires describing visual actions, which VLMs can do.
- **Known Hard Part:** Standard scene detection algorithms fail on slide-heavy lectures.



C2b: Video Translation & Dubbing

The Problem

- **The Core Issue:** Professional dubbing is too expensive, and subtitles split the viewer's attention.
- **Real Example:** Reading subtitles while trying to follow a fast, complex coding tutorial on screen reduces comprehension significantly.
- **Target User:** Education providers with international students.



Time	End time	Duration	Text	Original text
08.679	00:00:33.839	5.160	Именно так я обычно одет на судебных заседаниях.	This is how I would normally dress at court proceedings
04.399	00:00:37.399	3.000	И с ним сова Олли,	and with him is Ollie the Owl,
07.439	00:00:42.719	5.280	которая исполнена той мудрости, которую надлежит иметь юристу...	who is full of the kind of wisdom that lawyers are supposed to have
03.399	00:00:48.999	5.400	Так что, если я где-то ошибусь, я уверен, что Рампо и Олли поправят...	So if I make any mistakes, I'm sure that Rumpo and Ollie will correct me...
01.479	00:00:53.479	2.000	Если бы вы были жертвой преступления,	If you were a victim of a crime,
03.519	00:00:56.079	2.560	на что бы вы прежде всего обратили внимание и чего бы вы хотели?	where would you look first and what would you want?
07.879	00:01:01.999	4.120	Они хотят объяснения.	We are told that most victims of crime want closure. They want an explanation.
06.199	00:01:08.719	2.520	Они желают знать, почему это с ними случилось.	They want to know why it happened to them.
09.879	00:01:11.959	2.080	Но система правосудия для этого не предназначена,	The system isn't really made for that,
02.639	00:01:15.719	3.080	хотя есть косвенные механизмы,	although there are mechanisms around its fringes
06.319	00:01:19.919	3.600	позволяющие им получить какое-то понимание того, почему с ними произошло преступление.	which do allow them to get some kind of understanding why the crime happened to them.
00.479	00:01:25.759	5.280	По сути, английская система уголовного права, это система Англии и Уэльса	Essentially, in criminal law the English, that's the system for England and Wales
06.839	00:01:30.199	3.360	с похожими системами в Шотландии и Северной Ирландии,	with similar systems in Scotland and Northern Ireland,
00.239	00:01:33.919	3.680	которые, хотя немного отличаются от того, что я рассказываю сегодня,	though slightly different for everything that I'm going to say but broadly similar.
03.959	00:01:38.119	4.160	все же в целом похожие.	
08.159	00:01:39.759	1.600	Так вот, в Англии и Уэльсе у нас система состязательности в суде.	So, in England and Wales we have an adversary system.
06.479	00:01:49.479	3.000	Это означает, что обвинитель и обвиняемый или его адвокаты, в сущности, борются в суде	That means that the prosecutor and the defendant or the defendant's lawyers effectively do

Duration	Text	Chars/sec	Original text
2.520	Они желают знать, почему это с ними случилось.	17.86	They want to know why it happened to them.

Single line length: 17/28 Total length: 45 Single line length: 17/24

C2b: The Solution & Hard Parts

What to Build

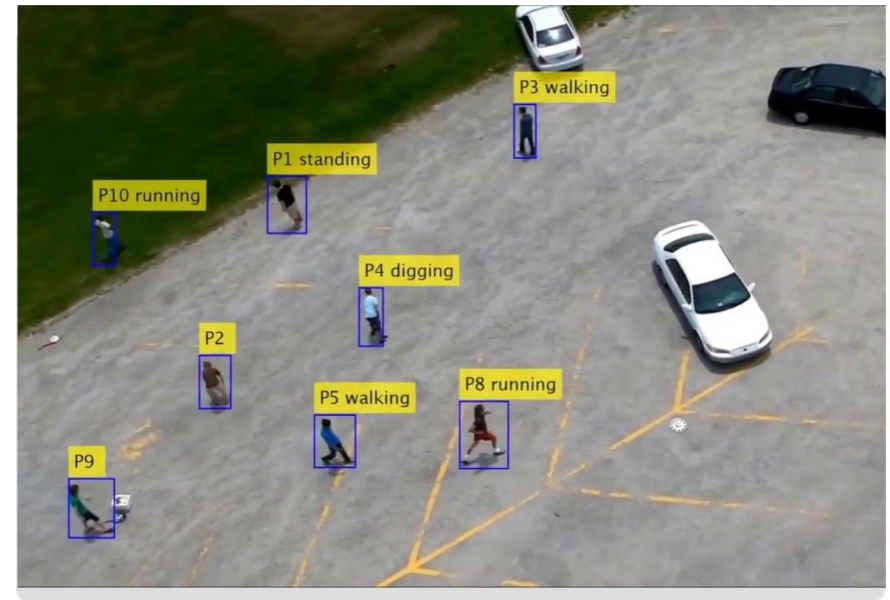
- **The Solution:** A full pipeline: extract audio → translate → synthesize speech → merge back into video.
- **Your Deliverable:** Compare your pipeline's cost vs. professional dubbing, with an honest quality assessment.
- **Known Hard Part:** Timing mismatches (languages take different times to speak) and compounding errors.

Text	Original text
Именно так я обычно одет на судебных заседаниях.	This is how I would normal
И с ним сова Олли,	and with him is Ollie the Ov
которая преисполнена той мудрости, которую надлежит иметь юрист...	who is full of the kind of wi
Так что, если я где-то ошибусь, я уверен, что Рампоу и Олли поправя...	So if I make any mistakes,
Если бы вы были жертвой преступления,	If you were a victim of a cr
на что бы вы прежде всего обратили внимание и чего бы вы хотели?	where would you look first-
Нан говорят, что большинство жертв преступлений хотят морального...	We are told that most victi
Они хотят объяснения.	They want an explanation.
Они желают знать, почему это с ними случилось.	They want to know w
Но система правосудия для этого не предназначена,	The system isn't real
хотя есть косвенные механизмы,	although there are mechan
позволяющие им получить какое-то понимание того,	which do allow them to get
почему с ними произошло преступление. По сути, английская система	why the crime happened to
уголовного права, это система Англии и Уэльса	the English, that's the syste
с похожими системами в Шотландии и Северной Ирландии,	with similar systems in Sco
которые, хотя немного отличаются от того, что я расскажу сегодня,	though slightly different for
все же в целом похожие.	but broadly similar.
Так вот, в Англии и Уэльсе у нас система состязательности в суде.	So, in England and Wales<
Это означает, что обвинитель	That means that the prosec
и обвиняемый или его адвокаты, в сущности, борются в суде	and the defendant or the de

C2c: Action Detection with VLM

The Problem

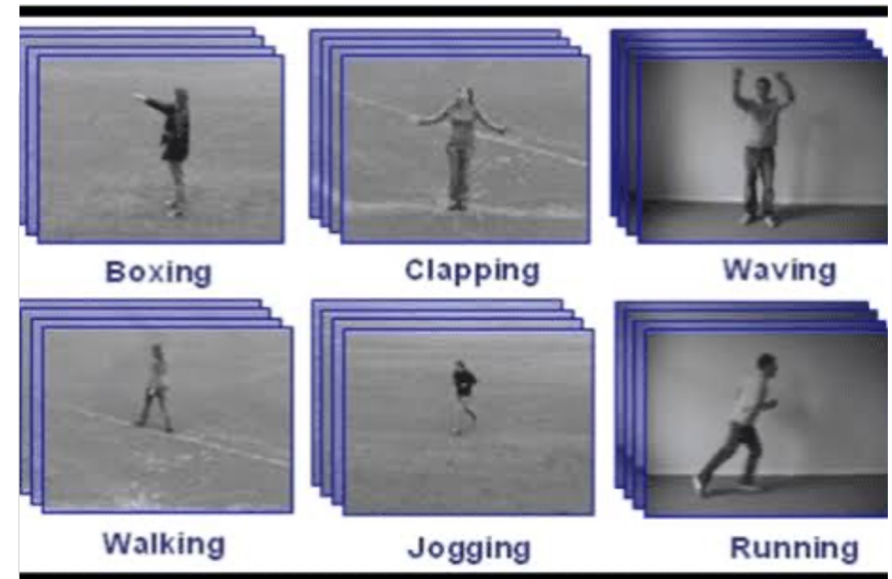
- **The Core Issue:** Verifying that a physical sequence happened correctly requires an expensive human to watch the video.
- **Real Example:** In a safety check or cooking procedure, doing the steps in the wrong order is a critical failure.
- **Target User:** Quality control functions or practical skills trainers.



C2c: The Solution & Hard Parts

What to Build

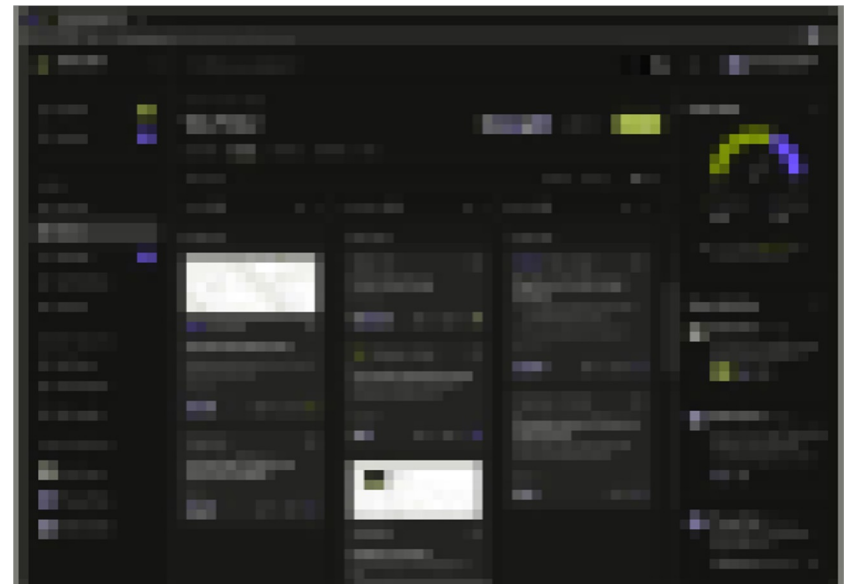
- **The Solution:** Detect actions and their order with timestamps across 20+ staged clips.
- **Your Deliverable:** Score detection, ordering, and counting separately.
- **Known Hard Part:** VLMs hallucinate plausibility — they guess the "usual" sequence instead of reporting what actually happened.



S1: Case Study & Portfolio Generator

The Problem

- **The Core Issue:** Companies do excellent work but fail to sell it because writing case studies takes unpaid hours.
- **Real Example:** You finish a great capstone project. When a recruiter asks for a portfolio, you have nothing ready to send because formatting takes too long.
- **Target User:** B2B service agencies or graduating students.



S1: The Solution & Hard Parts

What to Build

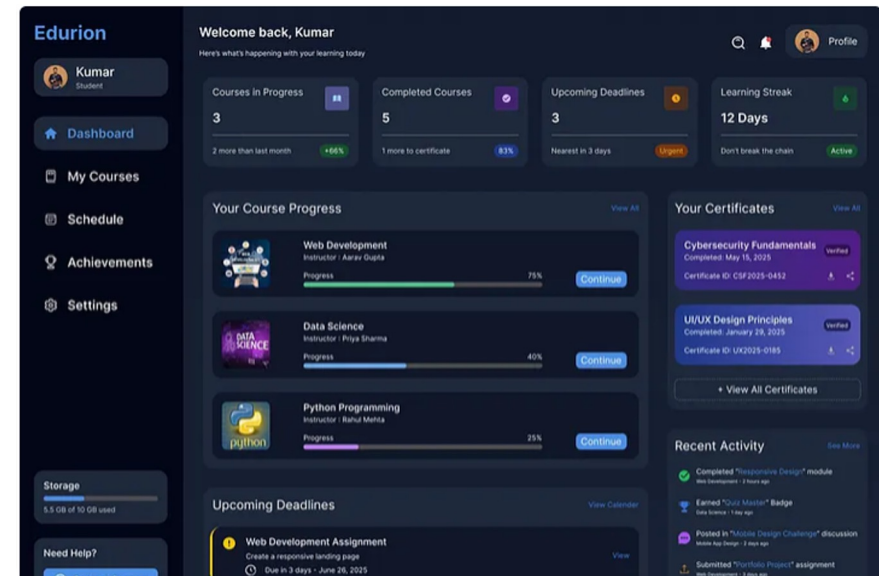
- **The Solution:** A tool that maps messy inputs (chat threads, rough notes) to a strict, professional PDF case study template.
- **Your Deliverable:** A system that includes a mandatory human review step before publishing.
- **Known Hard Part:** Preventing the AI from inventing plausible but fake metrics to sound better.



S2: Personalized Learning Roadmap

The Problem

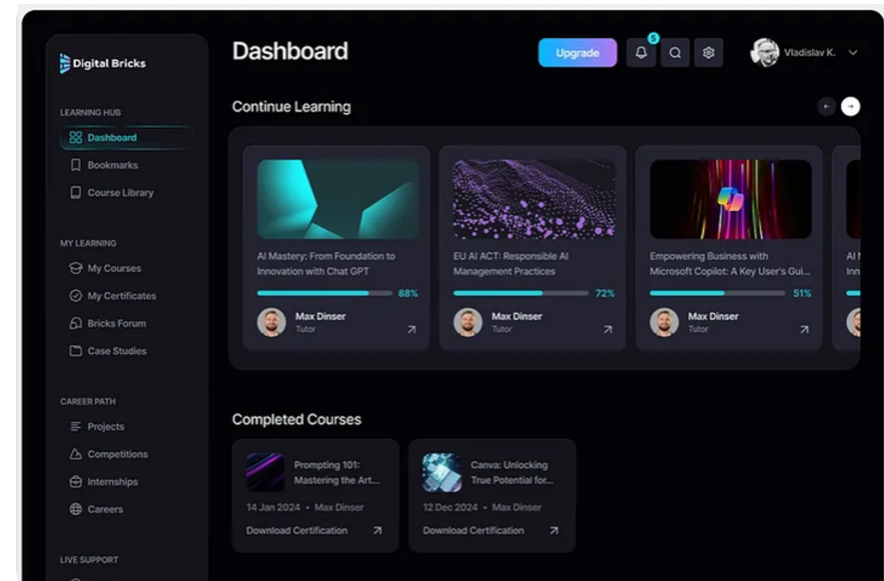
- **The Core Issue:** Free learning paths treat everyone identically, leading to high dropout rates.
- **Real Example:** An expert quits a tutorial from boredom because the first 3 weeks are basic, while a beginner quits from overwhelm.
- **Target User:** E-learning platforms (like roadmap.humblebee.ai).



S2: The Solution & Hard Parts

What to Build

- **The Solution:** Infer learner level via onboarding questions and generate a customized, ordered path.
- **Your Deliverable:** Produce visibly different paths for different learner profiles, explicitly justifying skipped material.
- **Known Hard Part:** Self-reported skills are unreliable, and writing good checkpoint questions is tough.



Tools & Platforms



Code & Models

Host your code on GitHub.
Use Hugging Face and
Unsloth for pulling and
fine-tuning models.



Compute & Services

Use Google Colab or
Google Cloud for compute.
Use ElevenLabs for high-
quality audio synthesis.



AI Assistants

You are encouraged to use
Claude, ChatGPT, or
OpenCode to accelerate
your development
process.

Next Steps

Review the problem pool documents carefully. Pay special attention to the "Known hard parts" before making your decision.

- **Week 2: Submit your 1st, 2nd, and 3rd choices with a written rationale.**
- **Team Formation: Form your teams (mix of IBT and ISE students).**
- **Submission: Submit your final role sheet to the instructors.**